

Predicting Breast Cancer Diagnosis Using Machine Learning

ISA 640: Capstone Project

Summer 2026

Project Overview

Breast cancer remains one of the most common cancers affecting women worldwide, and early, accurate diagnosis is critical to improving treatment outcomes. This project uses the Breast Cancer Wisconsin (Diagnostic) dataset: 569 samples with 30 numeric features derived from digitized images of fine needle aspirate (FNA) biopsies, describing characteristics of cell nuclei such as radius, texture, smoothness, and concavity, to build a machine learning classification model that distinguishes malignant from benign tumors. This project incorporates a graph database component (Neo4j), used to store, manage, and query the dataset as part of the broader data pipeline feeding into the machine learning workflow, from raw data ingestion through model-ready extraction.

Hypothesis

The central question this project seeks to answer is:

How accurately can supervised machine learning models predict whether a breast tissue sample is malignant or benign, based on diagnostic features extracted from digitized FNA images?

This leads to several supporting questions the project aims to address:

- Which cellular characteristics (e.g., radius, concavity, texture, symmetry) are the strongest predictors of malignancy?
- How does model performance compare across different supervised learning algorithms in terms of accuracy, precision, and recall?
- What are the practical implications of model performance for supporting clinical diagnostic decision-making?

Hypothesis: We hypothesize that ensemble-based models (e.g., Random Forest, Gradient Boosted Trees) and margin-based classifiers (e.g., Support Vector Machines) will outperform simpler linear models (e.g., Logistic Regression) on this dataset, given the non-linear relationships among cell nuclei measurements, while still maintaining strong interpretability through feature importance analysis.

Methodology

Task Type

This is a binary classification task. The target variable is the diagnosis label, encoded as malignant (M) or benign (B). Given the labeled nature of the dataset and the clear diagnostic outcome being predicted, this is a supervised learning classification problem rather than a prediction, ranking, or recommendation task.

Dataset

The Breast Cancer Wisconsin (Diagnostic) dataset (Wolberg, Street, & Mangasarian, 1995), sourced from the UCI Machine Learning Repository, contains 569 patient records and 32 variables: an ID field, a binary diagnosis label, and 30 continuous numerical features.

These features are grouped into three categories per cell nucleus measurement: mean, standard error (se), and "worst" (largest) value, across ten base measurements including radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension. The dataset contains no missing values, making it well-suited for supervised classification.

Data Pipeline

As part of the database requirement for this course, the dataset is loaded into Neo4j as the storage and query layer of the pipeline:

1. Data cleaning: the raw CSV is cleaned to remove formatting artifacts (e.g., trailing delimiter/empty column) prior to import.
2. Local import: the cleaned dataset is loaded into a local Neo4j Desktop instance using the Data Importer, with each record modeled as a "Tumor Diagnosis" node containing all diagnostic features as properties.
3. Cloud deployment: the local database is deployed to Neo4j Aura, providing a cloud-hosted, publicly reachable instance of the graph database.
4. Extraction: the data is queried from Aura via the Neo4j Python driver inside Google Colab and converted into a pandas DataFrame, ready for feature engineering and model training.

Planned Models (Tentative)

The following models will be evaluated and compared, consistent with approaches shown effective in prior research on this dataset:

- Logistic Regression: baseline linear classifier, valued for interpretability.
- Decision Tree: simple non-linear baseline.
- Random Forest: ensemble method expected to capture non-linear feature interactions with reduced overfitting risk.
- Gradient Boosted Trees (GBT): higher-capacity ensemble model, consistent with strong prior results on similar structured healthcare datasets.
- Support Vector Machine (SVM, RBF kernel): prior literature on this exact dataset reports this as a top-performing model (accuracy 96.8%), making it a strong candidate for comparison.